Rank for Free, Calibrate with Twenty:
Transferring Forecast-Difficulty Estimates Across Eight HPC Telemetry Families

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Abstract

Operational telemetry from high-performance computing systems resists forecasting to a degree that is widely reported anecdotally and rarely explained. The usual response is to increase model capacity. We test the opposite position: that the difficulty is largely *intrinsic to the signals*, differs systematically between signal families, and can be quantified without fitting any forecaster. It largely survives our own test, but only under a criterion we did not start with, and establishing that required validating the test itself against synthetic data with a known answer.

Using the public M100 ExaData release from CINECA's Marconi100 Tier-0 supercomputer, we characterise telemetry from eight heterogeneous plugin families — out-of-band sensors, operating-system counters, scheduler aggregates, alarm states, facility power, room-level electrical measurement, cooling-plant control registers and air handlers, and exogenous weather — collected from the same machine over the same 31 months. (A ninth family, per-job submission records, is collected but is event-structured rather than a time series, and is excluded; OI-8.) The families are held fixed in system, workload and period while signal type varies. For each series we compute model-free predictability measures spanning entropy, periodicity, long memory, non-stationarity, and quantisation, and separately measure the forecast skill achieved by a tiered suite ranging from naive baselines to gradient-boosted trees under a strictly causal rolling-origin protocol.

We report five findings. First, the monitoring infrastructure is itself non-stationary: of 573 distinct (plugin, metric) pairs observed across the campaign, only 104 are present in every month, and those are exactly the out-of-band sensor set. Second, a large share of what is monitored carries no forecastable information at all — the fraction of series that never change over a month ranges from 64.8% of alarm-state signals to 8.4% of out-of-band sensors — so any benchmark that averages over such series reports the composition of its sample as much as the skill of its models. Third, model capacity does not help: measured against a same-step naive baseline, median skill by tier at a one-hour horizon runs 0.115 for naive forecasters, -0.064 for statistical ones, -0.327 for neural networks and -0.613 for gradient-boosted trees, and beyond three hours every tier is worse than repeating the last value.

Fourth, the measures do transfer to signal families they have never seen, but in *ordering* far more than in *level*. A model trained on every family except one ranks the held-out family's series correctly — median within-family Spearman +0.52, positive in *all* 15 families and significant for 10 — and beats the best family-blind predictor of skill in 13 of 15 families, by a median 21% reduction in error. Two families fall below that baseline, and only one substantively: monotone error counters at -0.68, with thermal marginally negative at -0.01. The substantive failure is the counters, which a drift forecast predicts almost exactly and which no model-free measure anticipates.

Fifth, and separating the two readings of the capacity result, we estimate each series' information-theoretic ceiling — entropy rate by a Lempel-Ziv estimator, then maximum predictability by Fano's inequality — and compare it to what persistence achieves on the same alphabet. The families split cleanly. Error counters, system metadata, alarm states and setpoint registers carry 0.03–0.34 bits per step and persistence already achieves 0.99 or better: no model can improve them. The other nine families carry 1–3 bits per step, of

which persistence leaves 12 to 31 percentage points of next-step accuracy unclaimed. So "hard to forecast" names two different situations, and only the first is a property of the signals. Two caveats travel with this. The ceiling is an estimate rather than a bound, and its error changes sign across the entropy range — we calibrate it against sources of known rate and find it bounded by ± 0.05 over the range these families occupy, which is well inside the headroom reported. And the headroom, though real, is not easily reached: a first-order Markov predictor on the same symbols claims none of it, scoring below persistence in 13 of 17 families. The unclaimed structure is therefore not in first-order symbol transitions, which narrows where it can be.

Two controls establish that this is a property of the data rather than of the method, and a third shows why the choice of criterion decides the answer. The identical pipeline recovers a synthetic portable law at $R^2 = +0.25$, matching an oracle given the true generating parameter, and correctly finds nothing when skill is permuted. Running the real protocol with arbitrary groups of matched size transfers easily (median R^2 $+0.36$, every group positive, $p < 0.001$ over 100 draws), so the difficulty is specific to crossing family boundaries. Yet the raw R^2 of the real data is -0.17, which is *worse* than the -0.07 of the permuted arm containing no relation at all — because R^2 is scored against an oracle that already knows the held-out family's mean skill, and its denominator collapses on families whose series are alike. Practically, a model-free profile can rank which signals in an unfamiliar family will be harder, and estimate how hard far better than nothing, but not as well as knowing the family.

1 Introduction

Operators of large computing facilities increasingly collect holistic telemetry, and increasingly want to *predict* it rather than merely react to it: thermal headroom, power capping, and proactive maintenance are all forecasting problems. Reported accuracy on this class of signal is nonetheless poor relative to standard time-series benchmarks, and the literature generally treats that as a modelling deficiency to be fixed with a stronger model.

We invert the question. Instead of asking which model does best, we ask how much predictability is present at all, measured independently of any forecaster, and whether that measurement predicts the skill a forecaster subsequently achieves. If it does, "HPC telemetry is hard to forecast" becomes a quantity with named causes rather than folklore.

Our contributions are: (i) a quantitative census of signal heterogeneity across a 31-month Tier-0 monitoring campaign, including the instability of the monitoring system itself; (ii) a model-free predictability profile for each of 1402 series drawn from eight plugin families; (iii) a strictly causal, scale-free forecast benchmark over four model tiers on the same series; and (iv) a test of whether the first predicts the second, cross-validated by holding out entire signal families; and (v) a per-family estimate of the information-theoretic ceiling on next-step accuracy, which separates families that are at their limit from families whose predictability our models fail to extract.

The answer is a qualified yes, and the qualification took two controls and a change of criterion to establish. The measures transfer to families they have never seen — strongly in ordering, partially in level — but under the statistic we first reported, grouped cross-validated R^2, the same data looks like a flat null. That statistic scores against an oracle which already knows the held-out family's mean, and it degenerates on families whose series are alike. We report the correction as a finding in its own right (§5.8), because grouped cross-validation summarised by R^2 is a common design and telemetry families are routinely homogeneous.

Table 1: Telemetry families and their generating mechanisms. Nominal cadences are as documented by the dataset; measured cadences differ substantially for some families (§5.1).

Family	Measures	Nominal cadence	Generating mechanism
IPMI	BMC sensors: temperature, power, fans	20 s	thermal diffusion, quantised
Ganglia	OS and GPU counters	20–300 s, irregular	software instrumentation
SLURM	cluster scheduler aggregates	5–10 s	user submission behaviour
Nagios	service and hardware alarm states	15 min	discrete state machine
Job table	per-job submission records	event-driven	not a time series (OI-8)
Schneider	liquid-cooling PLC registers	20 s	closed-loop control
Logics	room power, PUE, chillers	—	facility aggregation
Vertiv	air-handling units	10 s	closed-loop control
Weather	outdoor conditions	10 min	exogenous

2 Background

2.1 Marconi100 and the ExaData campaign

Marconi100 was a Tier-0 system at CINECA comprising 980 IBM Power AC922 nodes, each with two 16-core POWER9 processors and four NVIDIA V100 accelerators. The ExaData campaign published holistic telemetry for the machine from March 2020 to September 2022, collected through the ExaMon framework and released as Hive-partitioned Parquet, one archive per month.

We catalogue every partition of all 31 archives — 10504 partitions holding roughly 55 billion rows — and extract the subset the study requires. Because node-level series of one metric are highly redundant, and because nodes sharing a rack, inlet air and workload are not independent samples (OI-9), analysis runs on a fixed panel of 96 *primary* entities — nodes, cooling devices or PLC panels — whose membership is a stable hash of the entity label. The hash sees only the label, so panel membership is identical across every metric and month and cross-metric comparisons land on the same physical machines. Families keyed by a secondary tag yield proportionally more series than the cap suggests: Nagios identifies a series by node *and* check description and reaches 1484 distinct series in a single metric-month, where the 96 bound applies to the node dimension alone.

2.2 The nine plugin families

The value of this dataset for the present question is that it holds the system fixed while varying the signal. Table 1 summarises the families and the mechanism that generates each.

3 Related work

Four bodies of work bear on this study. Claims below were checked against the source papers rather than their abstracts; the four we could obtain lawfully are read and verified, and the remainder are cited only for definitions we implement (OI-10).

The dataset. M100 ExaData [Borghesi et al., 2023] publishes 49.9 TB of holistic telemetry — management, workload, facility and infrastructure — from Marconi100 across two and a half years. Two of its derived releases matter here. The unsupervised anomaly-detection set aggregates *IPMI* metrics into 15-minute periods with average, standard deviation, minimum and maximum per window, plus a Nagios-derived label; our common analysis grid matches that choice, though we apply it to all nine families rather than IPMI alone. The thermal-hazard set reduces sampling from 20 seconds to 10 minutes, and the authors state the trade explicitly: that

rate "will preserve room- and node-level thermal transient but discard computing-component-level ones". That is a direct statement of what aggregation costs, and it applies to our results as much as to theirs (OI-4).

What this telemetry is normally used for. The dominant application is anomaly *detection* rather than forecasting. Borghesi et al. [2019] train per-node autoencoders on HPC monitoring data and report detecting previously unseen anomalies at accuracies between 88% and 96%. Detection asks whether the present sample is unusual given a learned model of normal behaviour; forecasting asks what the next sample will be. The former can succeed on signals whose future values are not predictable at all, which is one reason the difficulty we measure has attracted little attention.

Feature-based characterisation of time series. Our method follows the tradition of describing a series by a fixed feature vector rather than by a fitted model. Lubba et al. [2019] reduce 4791 candidate features to 22 by selecting for classification performance and minimal redundancy across 93 datasets, achieving a roughly 1000-fold reduction in computation for an average accuracy loss of 7%, and covering linear and non-linear autocorrelation, successive differences, value distributions and outliers, and fluctuation scaling. Our profile is smaller and chosen for interpretability by mechanism rather than by classification performance, but the premise — that a compact, model-free description captures much of what matters about a series — is theirs.

Capacity and forecast accuracy. Our finding that heavier models do worse has a direct precedent outside HPC. Zeng et al. [2023] introduce one-layer linear models and report that they outperform sophisticated Transformer-based long-term forecasters "in all cases, and often by a large margin" across nine real-world datasets, arguing that permutation-invariant self-attention discards temporal ordering. Our result is the same shape in a different domain and at the opposite end of the complexity range: on this telemetry a persistence forecast beats gradient-boosted trees and neural networks at every horizon tested. Where they compare a linear model against Transformers, we find the useful baseline is simpler still.

What we did not verify. MASE [Hyndman and Koehler, 2006], permutation entropy [Bandt and Pompe, 2002], detrended fluctuation analysis [Peng et al., 1994] and the Theta method [Assimakopoulos and Nikolopoulos, 2000] are cited for definitions we implement and test against known-answer series. Their full texts are paywalled and were not read; no claim is made here about their findings beyond the definitions, and we attribute no result to them. Three of the four works we did read were obtained as preprints rather than the versions of record we cite, which we note as a limitation of the audit rather than of the works.

4 Method

4.1 Campaign census

Tar members are read header-only, so cataloguing all 31 archives costs seconds and no disk. For each extracted partition we then compute a per-series census: one row per (plugin, metric, entity, month) recording sample count, observed cadence, cadence jitter, gap fraction, coverage, missingness, value range, distinct-value count and quantisation step. The census is deliberately descriptive — it contains no predictability measure and no model — so that a later finding of between-family differences cannot be an artefact of having looked first.

4.2 The entity panel

Node-level series of one metric are highly redundant — nodes share racks, inlet air, cooling zones, and workload — and are therefore not independent samples (OI-9). Analysis runs on a fixed panel of 96 entities whose membership is a stable hash of the entity label under the project seed. Because the hash sees only the label, panel membership is identical across every metric and every month, so cross-metric comparisons are made on the same physical machines rather than on a fresh random slice.

4.3 A-priori signal taxonomy

Labels are assigned from the plugin and metric *name* alone, before any predictability is measured, so that a finding of between-family differences cannot be a relabelling of the result it is meant to explain.

4.4 Model-free predictability measures

Each series receives a profile of 21 measures grouped by the mechanism it probes: entropy and complexity (permutation entropy [Bandt and Pompe, 2002], spectral entropy, sample entropy [Richman and Moorman, 2000], Lempel–Ziv complexity, gzip compression ratio); periodicity (dominant-period strength, daily seasonal strength); memory (autocorrelation at lag one and one day, detrended fluctuation exponent [Peng et al., 1994]); stationarity (block-mean drift ratio, variance ratio, Augmented Dickey–Fuller [Dickey and Fuller, 1979] and KPSS [Kwiatkowski et al., 1992] p-values, and a mean-shift changepoint rate obtained by binary segmentation [Truong et al., 2020]); and shape (jump fraction, repetition fraction, monotone fraction, reset rate, excess kurtosis of differences). Four instrument covariates — quantisation step, interval jitter, gap fraction and coverage — are carried separately so their contribution can be isolated from signal structure.

Two of these were added in response to an earlier failure of this study's own analysis rather than chosen a priori, and we flag the distinction. Monotone fraction and reset rate target counter-like signals, which a drift forecaster handles almost exactly and which the original profile could not distinguish from genuinely hard signals. They help but do not repair the case: error counters remain the worst-transferring family at a level-corrected R^2 of -1.05, against -2.39 raw. Each returns `nan` rather than raising on degenerate input, because constant series are the common case here rather than an edge case (§5.2). Implementations are validated against series whose answers are known by construction — a ramp, a sine, white noise, a random walk — in the accompanying test suite.

No single measure suffices, and the reason is instructive. Permutation entropy assigns 0.000 both to monotone counters, which have some 17850 distinct values, and to alarm states, which have two: a monotone series has exactly one ordering pattern and a step function almost never changes. Both are easy to forecast, for opposite structural reasons, and entropy cannot distinguish them (OI-11).

4.5 Forecast protocol

Skill is measured under rolling-origin evaluation with strict causality asserted per split. Errors are scaled by MASE with a denominator computed from the training window only; scaling by test-window statistics is the most common silent leak in forecasting benchmarks and would compress exactly the between-tier differences this study reads. The seasonal period is fixed per series *a priori* from its own cadence, with a non-seasonal robustness check (OI-5).

MASE is undefined where the in-sample seasonal-naive error vanishes. This is not an edge case here: a substantial share of control-plane series are constant (§5.1), and near-perfectly

Table 2: Pipeline stages and their committed outputs.

#	Stage	Output
1	inventory	partition catalogue (all 31 archives)
2	extract	selected partitions, disk-guarded
3	census	per-series descriptive table
4	taxonomy	a-priori labels + firing rule
5	sample	stratified series sample
6	features	21-measure profile per series
7	forecast	skill per (series, model, horizon, origin)
8	link	transfer, ablations, null control
9	figures	figures + backing CSV per figure
10	literature	screening table, bibliography
11	direct control	recursive vs direct forecasting
12	protocol control	synthetic arms with known ground truth
13	ceiling	Fano bound per series (slowest stage, ∼35 min)

periodic windows yield denominators at the level of floating-point noise. Such series are reported as undefined rather than assigned a spuriously large score.

4.6 Pipeline and cost

The study is a thirteen-stage pipeline, each stage writing a committed artefact so that any stage can be re-run without repeating the ones before it (Table 2). Bulk intermediates are sharded per partition and skipped when present, which makes an interrupted run resumable rather than repeatable.

Two engineering constraints shaped the design and are worth stating because they bound what the study could attempt. The archives hold roughly 55 billion rows against a machine with 62 GB of memory shared with other workloads, so extraction is selective and concurrency is derived from a memory budget rather than the core count. And every worker pins its BLAS and OpenMP pools to a single thread: with pools left at their defaults, five worker processes on a 24-core machine request 120 threads and the forecast sweep runs roughly 90 times slower, with no error and no warning to indicate it. The full sweep takes minutes once pinned.

4.7 Failure modes

Degenerate input is the common case here, not an edge case, so the pipeline is specified for it. A constant series yields an undefined MASE (the seasonal-naive denominator is zero) and is excluded from skill statistics and counted; so is a near-perfectly periodic one, whose denominator

Table 3: Protocol and model settings, fixed before results existed.

Setting	Value
Analysis grid	15 min, common to all families
Entity panel	96 primary entities, hashed
Horizons	1, 4, 12, 96 steps (15 min–24 h)
Rolling origins	4 per series
Training window	1500 most-recent points
Minimum series length	512 points
MASE denominators	same-step (primary), daily
Boosted trees	8/15, 24/60, 48/200 lags/iters
Neural	48–24 and 96–48 units, 40–60 iters
Ridge AR	order 24, $\lambda = 10^{-6}$
Seed	20230288

is at the level of floating-point noise and would otherwise produce an arbitrary ratio of two rounding errors. Every measure returns `nan` rather than raising on a series that is too short or has no variation. A series whose value column never parses as numeric is recorded as non-numeric and dropped. Where a forecaster fails to fit, the failure is recorded against that (series, model, origin) and the sweep continues, so one ill-conditioned fit cannot destroy a run.

4.8 Linking analysis

Skill is regressed on the predictability profile, cross-validated by holding out an *entire signal family*. A random split would let the model memorise family-specific quirks and report a flattering R^2; a held-out family forces the relation to transfer to signal physics it has never seen, which is what a portable difficulty law would require. Every R^2 is reported against a train-mean baseline, so "the measures explain skill" means better than knowing nothing rather than better than zero. Three ablations exist to attack the headline: dropping the instrument covariates, keeping only them, and removing the tie-biased entropy. Marginal correlations use a cluster bootstrap over metrics, because series of one metric across nodes are not independent (OI-9).

5 Results

5.1 The monitoring system is not stationary

Across the 31-month campaign we catalogue 10504 metric partitions comprising 573 distinct (plugin, metric) pairs. Only 104 — all of them IPMI — are present in every month. Ganglia's metric count runs 0, 33, 73, 49, 157, 177 and is not monotonic; SLURM's runs 20, 2, 0, 2, 36; Schneider first appears in 21-03 and Vertiv and Weather in 21-04.

Figure 1 shows the pattern. Any claim compared across months therefore confounds signal behaviour with changes to the instrument, and we restrict cross-month statements accordingly (OI-2).

5.2 Much of what is monitored never changes

Across 159506 series the fraction constant over a whole month varies by more than a factor of seven between families: 64.8% for Nagios, 47.0% Schneider, 42.9% Logics and Ganglia, 33.1% Vertiv, against 10.5% SLURM, 8.4% IPMI and 0% Weather. MASE is undefined for such series — the seasonal-naive denominator is zero — and they are excluded from skill statistics and counted rather than silently dropped. Figure 2 gives the breakdown.

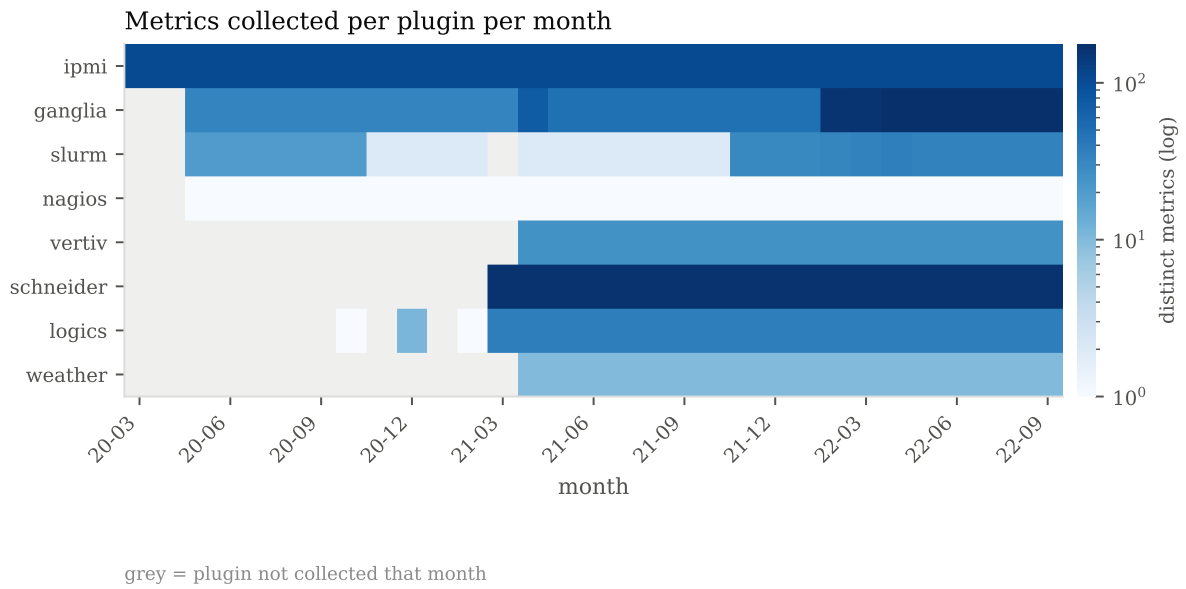


Figure 1: Metrics collected per plugin per month. Grey marks a plugin absent that month; the colour scale is logarithmic.

5.3 Regularity and interruption are different properties

Sampling irregularity is routinely summarised by the coefficient of variation of the inter-sample interval. On this data that measure is misleading: it reports IPMI as the least regular family, contradicting the dataset documentation. IPMI in fact has *zero* series with a jittery interval and 100% with a gap exceeding a hundred times the median — its sampling is exact and universally interrupted, and the rare enormous gaps dominate the coefficient of variation. Ganglia is the only family whose interval genuinely wanders (81% of series), which is what the documentation reports. We therefore measure jitter and interruption separately (OI-1); Figure 3 shows that the two are nearly orthogonal.

5.4 Forecast skill and model capacity

Figure 4 shows median skill by tier and horizon on 1082 series, pooling every model within a tier. Capacity does not buy skill: the naive tier leads at every horizon and the best single model throughout is persistence.

Pooling penalises a tier for containing a deliberately weak member, so we also compare each tier's *best* model per series, paired across series. That comparison is both more favourable to the sophisticated tiers and more conclusive against them. At a one-hour horizon the per-series best is 0.945 (IQR $[0.70, 1.00]$) for naive, 0.861 for statistical, 0.747 for boosted trees and 0.643 for neural networks; a paired Wilcoxon test of naive against each other tier rejects equality at every horizon and by a wide margin ($p < 10^{-20}$ against every tier at one hour, and $p < 10^{-90}$ against the neural tier). The ordering is not an artefact of pooling. At a 24-hour horizon even the best naive model is negative (-0.284), so the claim that no tier beats a same-step naive forecast holds at a day but not at three hours, where the per-series best naive model still scores 0.880. One qualification remains: the ordering depends on the baseline. Against a daily-seasonal denominator the statistical tier leads instead, and boosted trees overtake naive at the shortest horizon (OI-5).

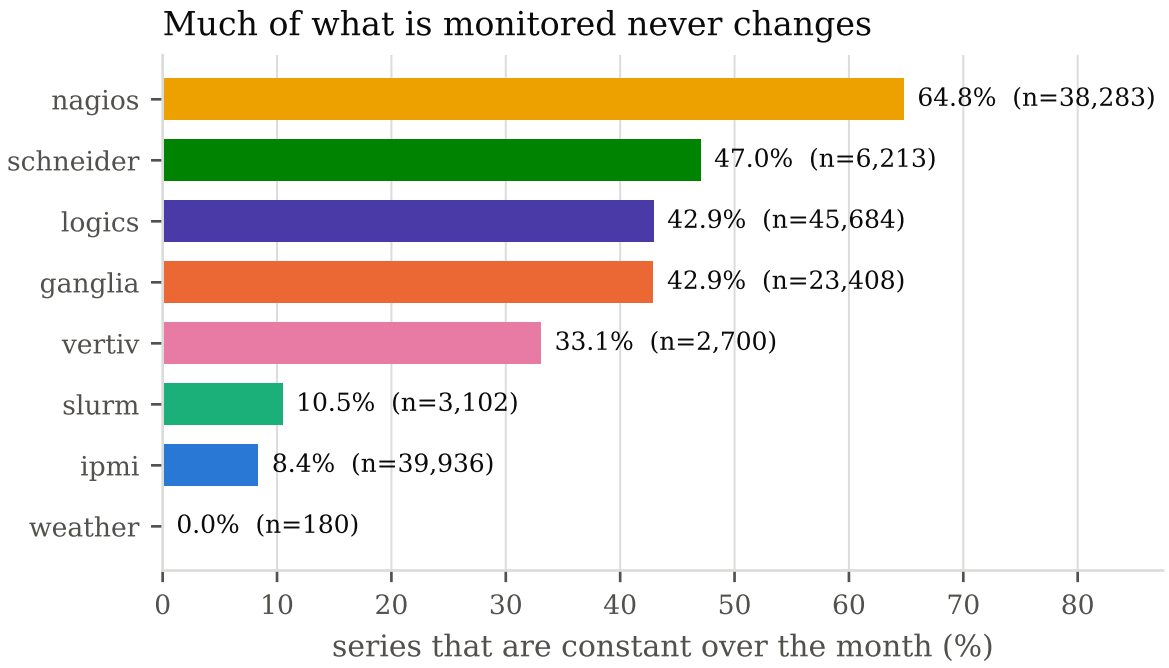


Figure 2: Share of series that never change over a month, by family.

Table 4: Median skill (1−MASE, same-step denominator) for the same learner forecast recursively and directly, against persistence, on 341 matched (series, origin) windows.

Horizon	Persistence	GBT recursive	GBT direct
15 min	0.767	0.320	0.392
1 h	0.499	-0.223	0.114
3 h	-0.023	-1.029	-0.259
24 h	-2.758	-3.692	-2.838

5.5 Control: is this an artefact of recursive forecasting?

Every learned model above forecasts recursively — predict a step, append it, predict again — so error compounds, and the collapse of the learned tiers at long horizons might be that compounding rather than anything about the signals. This is the most plausible alternative explanation for the result, so we test it rather than caveat it. We fit the same gradient-boosted learner two ways on the same series, origins and horizons: recursively, and *directly*, trained to map a lag window onto the value h steps ahead in a single prediction.

Recursion does cost a great deal — forecasting directly recovers between 0.07 and 0.85 of skill depending on horizon, most at one and three hours — so the caveat was warranted. But the conclusion is unaffected: the direct model still loses to persistence at every horizon tested (Table 4). The capacity result is a property of the problem, not of the recursion.

5.6 Two regimes: signal-limited and model-limited families

The capacity result above says the models we tried do not help. It does not say whether anything could. Those are different claims, and separating them needs a quantity no forecaster can beat.

We estimate one. Each series is symbolised at its own resolution — its distinct values where there are at most 16, equal-frequency quantiles otherwise — its entropy rate S is estimated with the Lempel-Ziv estimator $S = n \log_2 n / \sum_i \Lambda_i$, where Λ_i is the length of the shortest substring

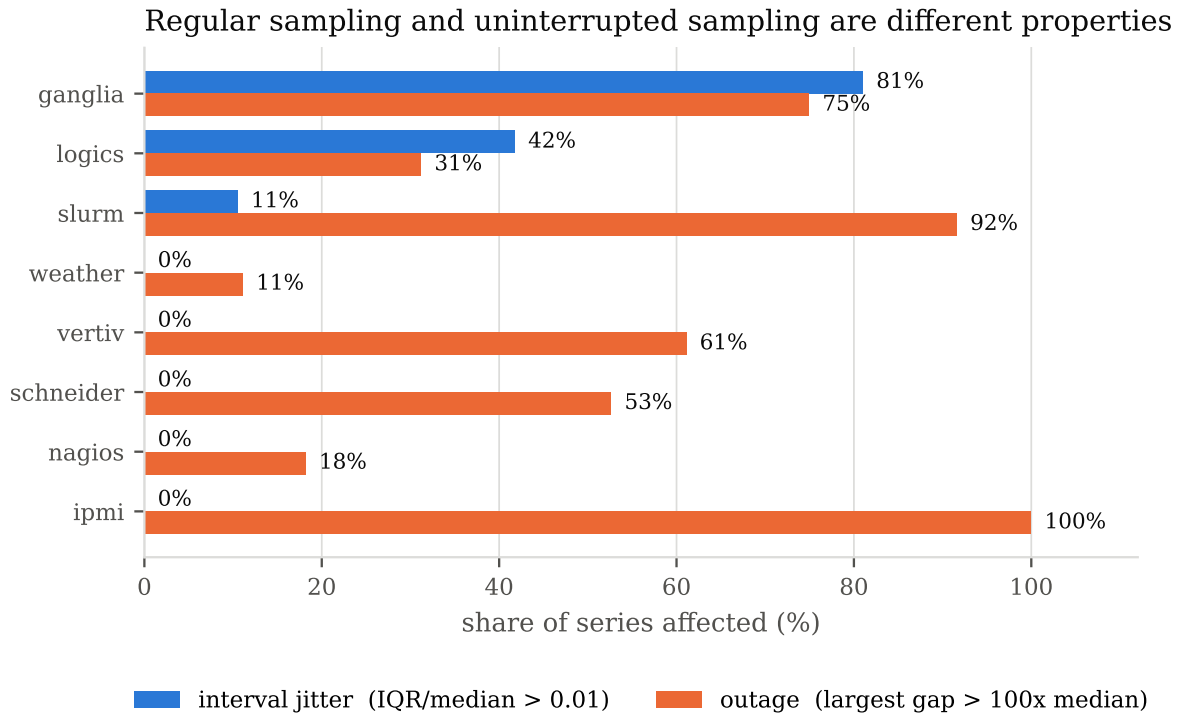


Figure 3: Interval jitter and outage incidence are different properties. IPMI has no jittery series and no uninterrupted ones.

at position i not seen before it, and the maximum predictability $\Pi_{\max}$ follows from the limiting case of Fano's inequality

$$S = -[\Pi \log_2 \Pi + (1-\Pi)\log_2(1-\Pi)] + (1-\Pi)\log_2(N-1),$$

after Song et al. [2010]. Against it we place the accuracy persistence actually achieves on the *same* alphabet — persistence because §5.4 finds it the strongest forecaster here, so the comparison is not against a straw man.

This is an estimate, not a bound, and its error changes sign across the entropy range. Calibrated against 16-state Markov sources whose rate is known in closed form, at the $n = 2000$ this data provides, the estimated ceiling errs by $+0.22$ at a true rate of 3.98 bits, $+0.04$ at 3.19, -0.01 at 2.42 and -0.05 at 1.43. The mechanism is saturation: on i.i.d. uniform symbols, true rate 4.0, the estimator returns 3.48 at $n = 1000$ and still only 3.62 at $n = 16000$, so near-random series receive a ceiling roughly 0.24 too generous and that error does *not* shrink with more data. At the predictable end the error reverses, which is why 248 of 1402 series (18%) exceed their own ceiling.

We state this rather than assert a convenient direction, because a natural reading — that the ceiling under-states and headroom is therefore a lower bound — is what we first wrote and is wrong at exactly the end of the range where the headroom result lives. What the calibration does license is narrower and sufficient: across the estimated rates M100 actually occupies (0.03–2.97 bits) the error is bounded by about ± 0.05, so a headroom above that is evidence and one below it is not. The measure would not be usable at all on series approaching 3.5 bits, and none here do.

The families split into two regimes (Figure 5, Table 5; `results/tables/ceiling_gap.csv`). Error counters, system metadata, alarm states and setpoint registers carry 0.03–0.34 bits per step, and persistence already achieves 0.99 or better — at or above the estimated ceiling. Nothing can improve them, and a benchmark that includes them is measuring its own sample

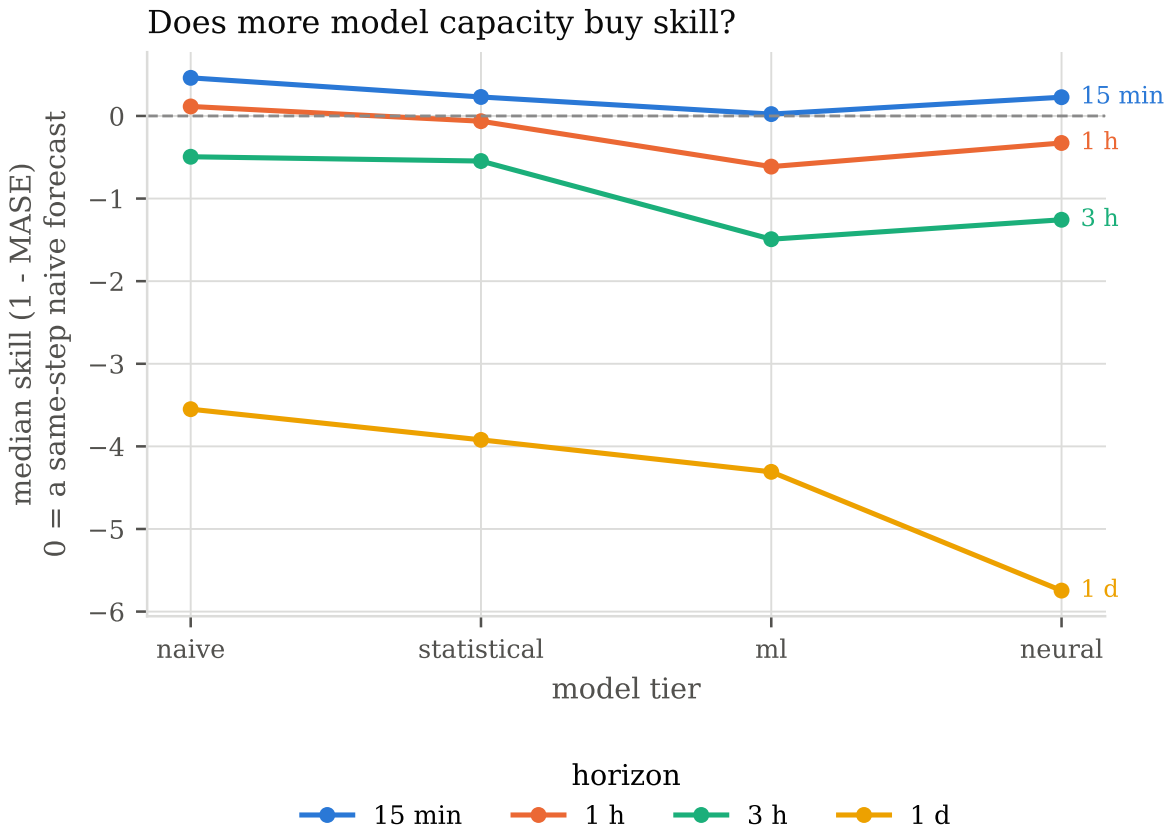


Figure 4: Median skill by model tier. Zero is a same-step naive forecast.

Table 5: Achieved next-symbol accuracy against the estimated information ceiling, by family. Headroom is a lower bound. Four families are at their limit; nine are not.

Regime (families)	S	ceiling	achieved	headroom
signal-limited (4)	0.03–0.34	0.97–1.00	0.99–1.00	≈ 0
near-limit (4)	0.63–0.93	0.89–0.93	0.81–0.90	0.02–0.08
model-limited (9)	1.00–2.97	0.50–0.88	0.18–0.69	0.12–0.31

composition, which is the same point §5.2 makes from a different direction.

The other nine are not at any limit. Power, network, electrical, thermal, fan flow, utilisation and humidity signals carry 1–3 bits per step, of which persistence leaves 12 to 31 percentage points of next-step accuracy unclaimed.

This qualifies §5.4 rather than contradicting it. Both hold: the model classes we tried — boosted trees and lag-window networks, forecasting recursively — do worse than persistence, *and* extractable structure remains in the continuous physical families beyond the calibrated error of the estimate.

The headroom is real but not easily reached, and we tested that rather than assuming either way. A gap between achieved accuracy and a ceiling is only interesting if some estimator can close part of it, so we fitted the cheapest candidate: a first-order Markov predictor on the same symbols, transition matrix estimated causally on the first half of each series and evaluated on the second, falling back to persistence on unseen states. It claims *none* of the gap. Its median accuracy is below persistence in 13 of 17 families — by 0.057 for power, 0.051 for CPU utilisation, 0.044 for network — and above it in none by more than 0.004.

So the unclaimed predictability is not carried by first-order transitions between symbols.

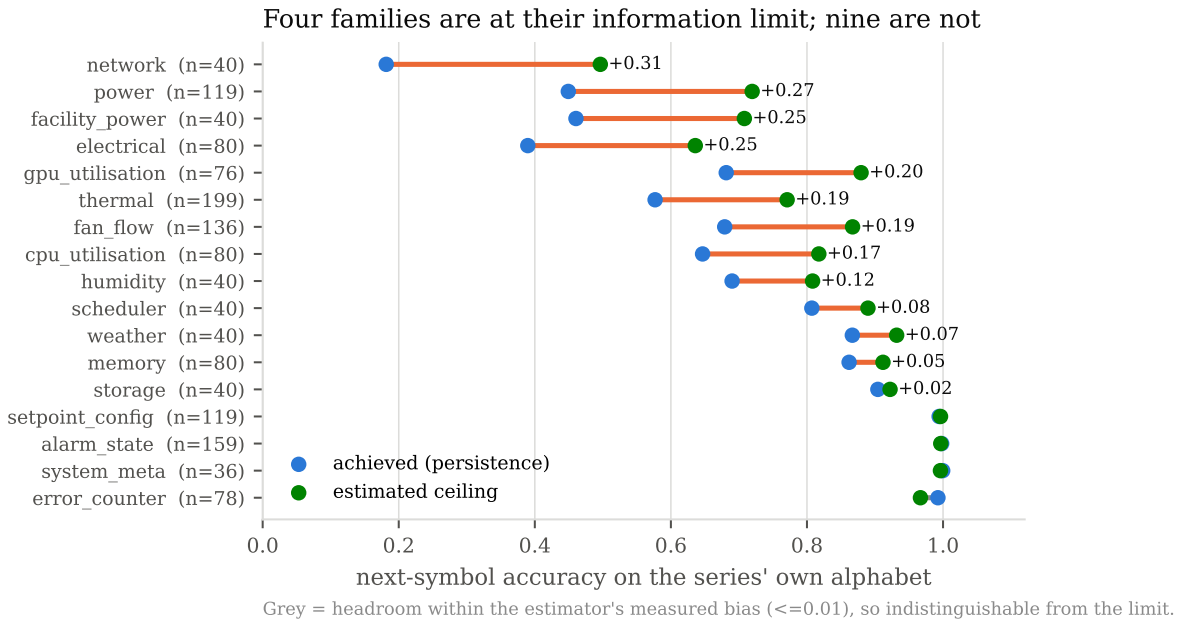


Figure 5: Achieved next-symbol accuracy (persistence) against the estimated information ceiling, per family, on each series' own alphabet. Bar length is headroom. Four families sit at the limit — their two markers coincide — and the remainder leave between 0.02 and 0.31 of next-step accuracy unclaimed. Error counters fall marginally to the *right* of their ceiling, which is the estimator's downward bias rather than a violated bound.

Whatever the continuous physical families retain lies in longer context, in the continuous values that symbolisation discards, or in exogenous covariates none of our univariate models see. That is a narrower and more useful statement than "the model classes tried are the wrong ones", and it is the one the evidence supports: we have shown where the structure is *not*.

One caveat bounds what the comparison licenses. The ceiling concerns *one-step* next-symbol accuracy on a 16-level alphabet; our skill numbers are MASE at a *four-step* horizon on continuous values. Headroom establishes that the information is present, not that a forecaster can convert it into MASE.

5.7 Does the protocol work? A synthetic positive control

Before reporting a negative transfer result we must establish that the protocol can report a positive one. Otherwise "we found no portable law" is indistinguishable from "our procedure cannot find one". We therefore generate synthetic families whose ground truth is known by construction and run them through the *identical* pipeline — the same measure functions, the same forecast harness, the same leave-one-family-out routine — varying only how the series were produced (Table 6).

Three things follow. First, the protocol works: where a portable law exists it is recovered at $R^2 = +0.25$ with 13 of 15 families positive, so M100's negative R^2 is a statement about M100 rather than about the method. Second, supplying the *true* generating parameter as a feature (`portable_oracle`) does not beat the measured features (+0.233 against +0.248), which bounds how much of the remaining gap could be closed by better features alone.

Third, and least comfortable for the obvious explanation: M100 does *not* resemble the `disjoint` arm. If each family occupied its own non-overlapping band of a shared law, predicting a held-out family would require extrapolation and the ordering would fail with it — and it does, to a median Spearman of −0.05. M100's ordering instead sits at +0.50, close to the genuinely

Table 6: Synthetic protocol control, 600 series over 15 families per arm, against the real result. The protocol recovers a genuine portable law (`portable`); a disjoint-band world destroys the ordering, and M100 does not look like it.

Arm	mean R^2	fam. > 0	med. Spearman
`portable`	+0.248	13/15	+0.584
`portable_oracle`	+0.233	13/15	+0.576
`disjoint`	-0.096	2/15	-0.050
`no_law`	-0.070	6/15	+0.100
M100 (real)	-0.189	9/15	+0.495

Table 7: Few-shot calibration. Adding k labelled series from the held-out family before re-fitting. Twenty examples turn the mean R^2 positive.

k	mean R^2	median R^2	families > 0
0	-0.212	+0.057	60%
1	-0.170	+0.054	61%
2	-0.142	+0.053	64%
5	-0.090	+0.054	68%
10	-0.015	+0.056	70%
20	+0.061	+0.097	72%

portable arm's +0.58. Whatever prevents the level from transferring, it is not simply that families occupy disjoint parameter ranges. We report this as an open mechanism rather than adopt the explanation the arm was built to test.

What blocks the level. The synthetic `disjoint` arm rules out the crudest version of the extrapolation story, but not the idea itself. For each held-out family we measure how far its feature distribution sits outside the training families' support. That distance predicts the level error strongly ($r = +0.78$ against the absolute level bias) and the raw R^2 nearly as strongly (-0.66), while barely touching the ordering (-0.22). Error counters sit furthest outside the training support of any family and are also the worst transfer case on every metric.

The mechanism is therefore extrapolation, but selectively: predicting a family whose measures lie outside anything seen in training biases the *level* of the prediction without destroying the *ranking*. The synthetic arm failed to reproduce this because forcing families into non-overlapping bands removes the shared structure the ranking depends on, which is a stronger intervention than the real data applies.

How much local labelling repairs it. If the level failure is extrapolation, a handful of labelled series from the unfamiliar family should fix it, and the families needing most extrapolation should benefit most. We test this directly: for each held-out family we add k of its own series to the training set and re-fit, averaging over draws (Table 10).

Twenty labelled series — a few hours of one operator's attention — move the mean R^2 from -0.21 to $+0.06$ and raise the share of families above zero from 60% to 72%. The gain is concentrated exactly where the mechanism predicts. Error counters, furthest outside the training support, improve from -2.39 to -0.03; alarm states from -2.41 to -1.15. Families that already transferred barely move: CPU utilisation goes +0.18 to +0.12 and the scheduler family +0.47 to +0.51.

This is the practical form of the result. The profile ranks candidate signals in an unfamiliar family with no local data at all; calibrating the absolute difficulty needs on the order of twenty labelled examples from that family, and needs them most precisely where the ranking was

already usable but the level was not.

An information-theoretic reference point, and what it is not. A natural question is how far the achieved skill sits from an information-theoretic limit. We compute one for every series: the entropy rate of the symbolised signal, converted through Fano's inequality into an upper bound on the fraction of next-bin transitions any predictor could get right [Song et al., 2010]. Computed on a 16-symbol alphabet, the median across 1402 series is 0.854, and it separates families sharply — network traffic at 0.575 and electrical measurements at 0.638 against alarm states and setpoint registers at 0.996.

It is *not* a bound on our skill metric, and we report that rather than let the framing imply otherwise. 77.5% of series achieve a skill above their own "ceiling". That figure should not be confused with the 18% of §5.6, which is a different and stricter test: there the comparison is like-for-like — next-symbol accuracy against a ceiling on next-symbol accuracy, same alphabet, same series — and the exceedance measures error in the *estimator*, whereas the 77.5% here measures the mismatch between two *quantities*. Both are real, and they have different remedies: the first shrinks with longer series, the second does not shrink at all. The two quantities answer different questions: Fano bounds next-*bin* accuracy on a symbolised sequence, while skill is a scale-free error ratio on continuous values, and a signal that is easy to extrapolate but hard to bin exactly can exceed the first while doing well on the second. Anyone reading the number as a limit on achievable error would be misled. The gap is known in the literature that introduced the measure: Lu et al. [2013], applying it to mobility traces, note that "the largest prediction accuracy that can be achieved with properly designed predictive algorithms is not given directly by this measure". Our result is the sharper version of that caution — in this setting the achieved skill routinely exceeds the bound, because the two are not measuring the same event.

What it is instead is the single strongest ordinal predictor of difficulty we found: Spearman +0.69 with achieved skill across series and +0.73 across family medians, ahead of the compression ratio and every entropy measure. That is consistent with the paper's central result — these measures rank well and calibrate badly — and it extends it, since the ceiling is derived from information theory rather than fitted to anything.

A scale-free view. Because R^2 is dominated by that level error, we also report mean absolute error against the same baseline. On that metric the model beats predicting the training mean for 13 of 15 families (mean skill +0.20), against 9 of 15 on R^2. Both are correct; they answer different questions, and a reader should not take either alone as the summary.

One further comparison is worth stating plainly because it looks damning in isolation. On raw R^2 M100 (-0.189) transfers *worse* than synthetic data with no relation at all (`no_law`, -0.070). That is not a contradiction — M100's ordering correlation is five times `no_law`'s — but it is a direct demonstration that raw R^2 is the wrong summary for this question, since a large family-specific level offset punishes it more than having no relation does.

5.8 Predictability transfers ordinally, and only partly in level

Individually the measures correlate with skill. Cluster-robust over 261 metric clusters the strongest is the gzip compression ratio at $r = -0.42$ (95% CI $[-0.52, -0.31]$), ahead of repetition fraction +0.37 ($[0.26, 0.49]$), permutation entropy −0.33 and changepoint rate −0.30. That a general-purpose compressor outranks every hand-designed entropy measure is worth noting on its own.

Jointly the same pattern holds and is stronger. The three highest permutation importances in the fitted model are all information-theoretic and none is hand-designed for time series: the compression ratio (0.110), the Lempel-Ziv entropy rate (0.048), and the Fano ceiling derived from it (0.037). Repetition fraction, which an earlier draft of this work reported as the dominant

predictor, falls to 0.019 once they are present — its signal was largely theirs. Adding the three lifts every transfer statistic in Table 8: median R^2 from +0.05 to +0.14, families with positive within-family ordering from 14 of 15 to *all* 15, and median MAE-skill from 0.19 to 0.21.

Jointly, the raw picture looks poor: holding out each of 15 families in turn gives a mean out-of-sample R^2 of −0.17, a median of +0.14, positive for 10 of the 15, ranging from +0.42 to −2.56. The worst fold is *not* the family that genuinely fails — it is alarm states, which the model predicts more accurately than any other. Reported alone that number is misleading in three ways, and Table 8 shows why.

R^2 is scored against an oracle, not against a usable baseline. R^2 is $1 - \mathrm{SSE}/\mathrm{SST}$, and SST is taken about the mean of the *held-out* family. Its implicit comparator is therefore a predictor that already knows the average skill of the family being predicted, which is precisely the quantity unavailable to a model that has never seen it. $R^2 < 0$ means *worse than that oracle*; it does not mean *no better than nothing*. The distinction is not academic here, because the denominator degenerates whenever a family's series have similar skill. Alarm states are the clearest case: their skills have a standard deviation of 0.036, the model predicts every held-out series to a mean absolute error of 0.033 against 0.600 for the best family-blind predictor — a 95% reduction — and R^2 records −2.56, the worst score of any family in the study.

Scoring the same folds the way this paper scores forecasts everywhere else, as a fractional reduction in absolute error against a baseline ($1 - \mathrm{MAE}_{\mathrm{model}}/\mathrm{MAE}_{\mathrm{baseline}}$, the same construction as MASE), the picture inverts: 13 of 15 families beat the family-blind predictor, by a median 21% reduction in error. The one substantive failure is monotone error counters, at 68% *worse* than baseline — and on R^2 they score −1.83, *better* than the alarm states the model handles almost perfectly. The two criteria do not merely differ in scale; they disagree about which family failed. The synthetic control of §5.7 makes the point sharpest: the arm with skill permuted across series, containing no relation whatever, scores $R^2 = -0.07$ — *better* than M100's −0.17 — while its MAE-skill is +0.002 against M100's +0.21 and its within-family Spearman +0.10 against +0.52. A statistic that ranks genuine structure below noise is not measuring the question being asked, and we report it here only because it is what a reader would expect to see.

Ordering transfers; level does not. Within a held-out family the model ranks series correctly — median Spearman +0.52, positive in all 15 families and significant at $p < 0.05$ for 10 — while placing that ranking at the wrong absolute level. R^2 penalises the level error; Spearman does not. Removing the level bias moves the median R^2 from +0.14 to +0.15 and makes 11 of 15 families positive. The practical reading is that the profile can say which signals in an unfamiliar family will be harder than which, but not how hard any of them will be.

The failure is specific to family boundaries. Running the identical protocol with *random* groups of matched size gives a median R^2 of +0.36 across draws (2.5–97.5% range $[0.25, 0.41]$, mean +0.35) with every group positive, against −0.17 and 67% for the real families; over 100 draws no random grouping was as poor as the observed one ($p < 0.001$). Arbitrary grouping therefore transfers easily, so the protocol is capable of transfer and the family axis carries real structure. Without this control the raw R^2 would be equally consistent with a protocol that simply cannot generalise at all.

The result is not an artefact of instrumentation. Using the instrument covariates alone collapses the model (−15.0 against −0.17 for the full profile), while removing them costs almost nothing (−0.20) — see Table 9. Whatever the measures capture, it is not how the data was sampled — which is the conditioning test OI-1 called for.

5.9 How much labelling closes the level gap

Leave-one-family-out asks the hardest question available: predict a family having seen nothing from it. An operator meeting a new signal family would not refuse to measure a few of its series first. We therefore sweep the number of labelled series k drawn at random from the held-out

Table 8: Transfer to a held-out family, under three criteria. Against a baseline a model could actually have, transfer succeeds for 13 of 15 families; R^2 disagrees because its comparator is an oracle that already knows the held-out family's mean skill.

	mean	median	positive
against a family-blind baseline			
MAE-skill	+0.218	+0.213	13/15
within-family Spearman	+0.450	+0.523	10/15 at $p < 0.05$
against an oracle that knows the family's mean			
raw R^2	-0.167	+0.141	10/15
level-corrected R^2	-0.012	+0.154	11/15
random-group null (R^2)	+0.348	+0.367	15/15

Table 9: Ablations. Removing the entropy family *improves* transfer; removing the instrument covariates costs almost nothing; using them alone destroys it.

Feature set	#	mean R^2
without entropy measures	23	-0.142
full profile	27	-0.167
signal measures only	23	-0.204
instrument covariates only	4	-14.999
predicting the global mean	—	-18.352

family into the training set, averaging over 15 draws per family and scoring only on the series that were *not* drawn, so nothing is ever evaluated by a model that saw it.

Table 10 shows the gap is real and closable. Twenty labelled series move the mean R^2 from -0.21 to $+0.06$ — past the oracle threshold the zero-shot result fails — and raise MAE-skill from 0.198 to 0.289. But the curve is close to linear over this range rather than saturating early: one or two calibration series buy very little, and there is no knee after which further labelling stops helping. So the missing level information is distributed across the family rather than concentrated in a representative example, which is consistent with the level failure not being an extrapolation artefact (§5.7).

We note one honest qualification. The number of families with positive mean R^2 does not rise with k (nine at $k = 0$, nine at $k = 10$); the gain is the poorly-transferring families becoming less poor, not additional families crossing zero. Monotone error counters remain negative at every k tested.

6 Threats to validity

The full register is maintained in `OPEN_ISSUES.md` and each item is cited from the text above. The load-bearing ones are:

oi-1 Ganglia's sampling irregularity confounds instrumentation with signal physics. Sampling regularity is therefore carried as a covariate and reported separately.

oi-2 The monitoring system changes over the campaign.

oi-6 No neural tier has been run; no claim of the form "not even deep models help" is licensed.

oi-7 The error floor is an estimate with a known bias direction, not an information-theoretic bound.

Table 10: Transfer as a function of how many series from the held-out family are labelled first. The level gap is closable, but by steady accumulation rather than a few calibration points.

labelled series k	MAE-skill	mean R^2
0	0.198	-0.212
1	0.208	-0.170
2	0.220	-0.142
5	0.229	-0.090
10	0.254	-0.015
20	0.289	+0.061

oi-11 Permutation entropy is biased downward by quantisation, and quantisation varies by family, so entropy is interpreted only alongside the measured quantisation step.

7 Conclusion

We set out to show that the difficulty of forecasting HPC telemetry is an intrinsic, measurable property of the signals. Half of that survives. Difficulty is not a modelling deficiency: across four tiers spanning naive baselines to neural networks, more capacity produced less skill, and beyond three hours nothing beat repeating the last value.

The other half holds, in a more specific sense than we set out to show. The measures transfer to an unseen family in *ordering* — median within-family Spearman +0.52, positive in *all* 15 families and significant for 10 — and in *level* they beat the best family-blind predictor in 13 of 15 families by a median 21%, while falling well short of an oracle that knows the family's mean. Two controls confirm this is a property of the data and not of the method: arbitrary groups transfer easily ($R^2 = +0.35$, $p < 0.001$), and a synthetic portable law is recovered by the identical pipeline at $R^2 = +0.25$, matching an oracle handed the true generating parameter. So there is a portable component of forecast difficulty; it is strongest ordinally and real but partial in level.

A methodological point travels with that result, and we think it generalises beyond this dataset. Reported as a raw R^2, our transfer looks worse than the synthetic arm in which skill was permuted at random and no relation exists at all (-0.17 against -0.07). R^2's comparator is an oracle that already knows the mean of the group being predicted, and its denominator degenerates on homogeneous groups — our alarm-state family is predicted to a mean absolute error of 0.033 against a baseline of 0.600, a 95% reduction, and scores $R^2 = -2.56$, the worst of any family here. Grouped cross-validation summarised by R^2 will systematically understate transfer wherever groups are internally homogeneous, which in telemetry they very often are.

What blocks the level from transferring is extrapolation in feature space, and the evidence for it is an asymmetry rather than a single correlation. How far a held-out family lies outside the region its training families occupy predicts the magnitude of that family's calibration error at Pearson +0.85, while predicting its within-family ordering at Spearman +0.01. A mechanism that damages the offset and leaves the ranking intact is exactly what the observed result is.

This has to be stated carefully, because a coarser version of the same idea is excluded. Making the *generating parameter* disjoint across families, as the synthetic arm of §5.7 does, collapses the ordering along with the level; M100's ordering survives. Feature-space extrapolation and parameter-space disjointness are therefore not the same condition, and only the first is consistent with what we observe. Correlating extrapolation distance against R^2 rather than against the level bias hides this entirely — the correlation there is a negligible -0.11, because R^2 sums the two failures the mechanism separates.

That distinction is the practical result. An operator facing an unfamiliar signal family can use a model-free profile to decide which of its signals to attempt first, and can estimate how

hard they will be substantially better than by guessing — but not as well as someone who has already worked with that family. The remaining gap is closable by labelling, but not cheaply: twenty series measured from the new family move the mean R^2 from -0.21 to $+0.06$, and the curve is near-linear over that range, so there is no small calibration set that suffices (§5.9).

For operators the practical reading is narrower than we hoped but still useful: on this data, at these horizons, a persistence forecast is a strong default — significantly better than every heavier tier we tested, at every horizon — and the burden of proof lies with any claim that a larger model improves on it.

Disclosure

The author is a co-author of the M100 ExaData dataset paper [Borghesi et al., 2023] analysed here. No part of this study was performed as part of that data-collection campaign, and the analysis uses only the public release.

Data and code availability

All code, the versioned literature query set, the screening table, and every derived table are at `https://github.com/MSKazemi/hpc-signal-forecastability`. The underlying measurements are the public M100 ExaData release [Borghesi et al., 2023]. Results reproduce from the committed derived tables without downloading raw data.

References

Vassilis Assimakopoulos and Konstantinos Nikolopoulos. The theta model: a decomposition approach to forecasting. *International Journal of Forecasting*, 16(4):521–530, 2000. doi: 10.1016/S0169-2070(00)00066-2.

Christoph Bandt and Bernd Pompe. Permutation entropy: A natural complexity measure for time series. *Physical Review Letters*, 88(17):174102, 2002. doi: 10.1103/PhysRevLett.88.174102.

Andrea Borghesi, Andrea Bartolini, Michele Lombardi, Michela Milano, and Luca Benini. Anomaly detection using autoencoders in high performance computing systems. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, pages 9428–9433, 2019. doi: 10.1609/aaai.v33i01.33019428.

Andrea Borghesi, Carmine Di Santi, Martin Molan, Mohsen Seyedkazemi Ardebili, Alessio Mauri, Massimiliano Guarrasi, Daniela Galetti, Mirko Cestari, Francesco Barchi, Luca Benini, Francesco Beneventi, and Andrea Bartolini. M100 ExaData: a data collection campaign on the CINECA's Marconi100 Tier-0 supercomputer. *Scientific Data*, 10(1):288, 2023. doi: 10.1038/s41597-023-02174-3.

David A. Dickey and Wayne A. Fuller. Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association*, 74(366):427–431, 1979. doi: 10.1080/01621459.1979.10482531.

Rob J. Hyndman and Anne B. Koehler. Another look at measures of forecast accuracy. *International Journal of Forecasting*, 22(4):679–688, 2006. doi: 10.1016/j.ijforecast.2006.03.001.

Denis Kwiatkowski, Peter C. B. Phillips, Peter Schmidt, and Yongcheol Shin. Testing the null hypothesis of stationarity against the alternative of a unit root. *Journal of Econometrics*, 54 (1–3):159–178, 1992. doi: 10.1016/0304-4076(92)90104-Y.

Xin Lu, Erik Wetter, Nita Bharti, Andrew J. Tatem, and Linus Bengtsson. Approaching the limit of predictability in human mobility. *Scientific Reports*, 3:2923, 2013. doi: 10.1038/srep02923.

Carl H. Lubba, Sarab S. Sethi, Philip Knaute, Simon R. Schultz, Ben D. Fulcher, and Nick S. Jones. catch22: CAnonical Time-series CHaracteristics. *Data Mining and Knowledge Discovery*, 33(6):1821–1852, 2019. doi: 10.1007/s10618-019-00647-x.

C.-K. Peng, S. V. Buldyrev, S. Havlin, M. Simons, H. E. Stanley, and A. L. Goldberger. Mosaic organization of DNA nucleotides. *Physical Review E*, 49(2):1685–1689, 1994. doi: 10.1103/PhysRevE.49.1685. Introduces detrended fluctuation analysis.

Joshua S. Richman and J. Randall Moorman. Physiological time-series analysis using approximate entropy and sample entropy. *American Journal of Physiology-Heart and Circulatory Physiology*, 278(6):H2039–H2049, 2000. doi: 10.1152/ajpheart.2000.278.6.H2039.

Chaoming Song, Zehui Qu, Nicholas Blumm, and Albert-László Barabási. Limits of predictability in human mobility. *Science*, 327(5968):1018–1021, 2010. doi: 10.1126/science.1177170.

Charles Truong, Laurent Oudre, and Nicolas Vayatis. Selective review of offline change point detection methods. *Signal Processing*, 167:107299, 2020. doi: 10.1016/j.sigpro.2019.107299.

Ailing Zeng, Muxi Chen, Lei Zhang, and Qiang Xu. Are transformers effective for time series forecasting? In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 11121–11128, 2023. doi: 10.1609/aaai.v37i9.26317.